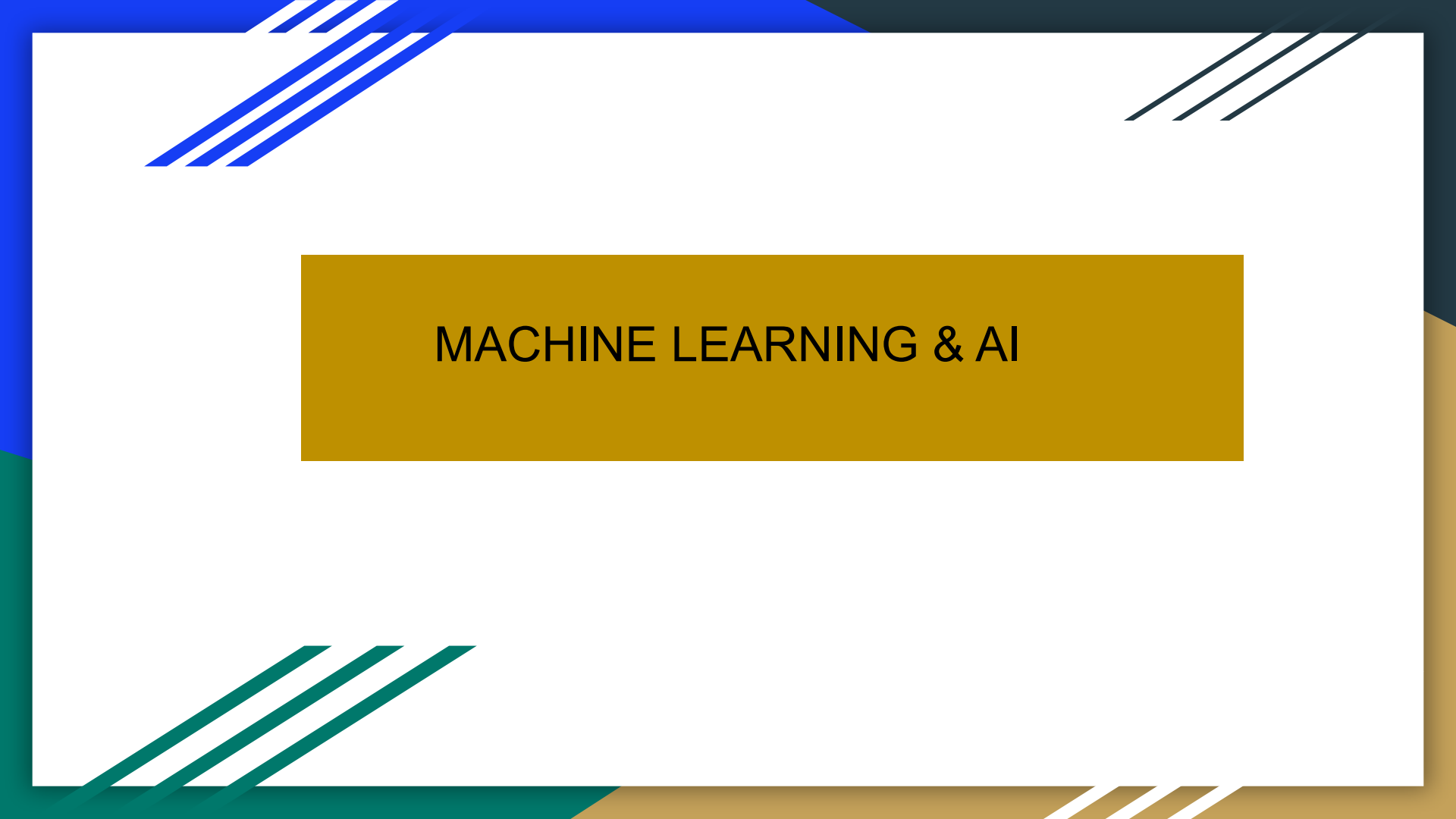




MACHINE LEARNING & AI

WEEK 3 – Unsupervised Learning + Model Evaluation

 **Focus: Clustering, Dimensionality Reduction, Scaling**

 **Topics Covered:**

- K-Means Clustering
- PCA (Principal Component Analysis)
- Elbow method for choosing K
- Model saving/loading (joblib, pickle)

Assignments:

1. Perform K-Means on Iris dataset and visualize clusters.
2. Apply PCA to reduce dataset dimensions.

Mini Project 3: **“Iris Flower Clustering Project”**

Dataset:

 [Iris Dataset \(UCI / Kaggle\)](#)

Tasks:

- Apply K-Means ($k=3$)
- Visualize clusters (scatter plot)
- Compare predicted clusters vs true labels

 *Skill Gain:* Unsupervised ML + Visualization.